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Course 608

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Research Proposal

Title

10 ‘Using specific machine learning on personal genomic data to predict specific warfarin dosage in the South Asian population.’

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Using specific machine learning on personal genomic data to predict specific warfarin dosage in the South Asian population

1. Introduction

Warfarin is the most widely used blood thinner. It is an anticoagulant that slows the body's process of making clots by targeting specific clotting factors. Synthesized in 1948, this drug was first introduced as a potential rat poison[1]. It was approved for human use in 1954 under the name Coumadin. Since then warfarin has been prescribed for atrial fibrillation (AFib) and thromboembolic disorders like Venous thromboembolism (VTE) . As the name suggests it is a critical condition developed from the formation of clot inside a vein leading to the lungs.

Vit K which is produced by the liver is the main culprit behind the synthesis of blood clots leading to strokes and heart attacks. Warfarin works by interfering with the production of VitK. Vit K can exist in different forms. For example, one specific form is the Quinone which stays in an oxidised state. With the aid of NADPH , the quinone is reduced to form VitK Hydroquinone . In the next step, VitK Hydraquinone drops off its electrons into Gamma Glutamyl Carboxylase and gets oxidised into VitK epoxide. Finally, through the enzyme VitK Epoxide Reductase , VitK epoxide gets reconverted to its original form VitK Quinone.[2] Thereby, Vitamin K is required for the synthesis of prothrombin, Factor VII, Factor IX, Factor X, involved in the clot creation process.[3] When Warfarin crosses the digestive tract and reaches the liver the cycle breaks. Warfarin inhibits the enzyme VitK Epoxide Reductase, so VitK Quinone cannot be produced. Hence, the Gamma Glutamyl Carboxylase cannot add the carboxyl group to the Factor Proteins so they remain dysfunctional.

2. Background To The Research Proposal

Warfarin's pharmacokinetic and pharmacodynamic pathways entail more than 30 genes influencing the formation of VitK and its hepatic uptake. Warfarin's molecular target is encoded by the gene *VKORC1* and the metabolisation of the enantiomer of warfarin is done by *CYP2C9*. Due to slight differences in Minor Allele Frequencies in these alleles, the resulting key single nucleotide change varies from population to population. Consequently, complications arose due to specificity of the drug dosage among older US adults between 2007 and 2009. This had led to 33% of an estimated 99,628 annual hospitalizations. [4]

3. Research Objectives

1. To predict the warfarin maintenance dose in South Asian population using the machine learning approach Random Forest Regression(RFR)

4. Research Aim

1. To compare between gene *CYP2C9* and *VKORC* using Random Forest Regression (RFF) in figuring out the most efficient methods for gene analysis to predict warfarin dosage.

5. Literature Review

A six month long study was conducted to identify the warfarin induced in the Cardiology Unit of a teaching hospital in India. The duration of this clinical trial lasted 6 months. The variables were age, gender, length of stay, number of drugs dispensed, frequency of bleeding and drugs involved in the bleeding.The number of people who fell into each category was counted and expressed as percentage. The chi squared test was used to find association with each sector. A total of 235 patients were included and the results showed 61 people had faced warfarin induced bleeding. The limitations to this methodology include the inconsideration of the gene *CYP2C9* and *VKORC*. As previously stated, warfarin's molecular target is encoded by the gene *VKORC1* and the metabolisation of the enantiomer of warfarin is done by *CYP2C9*. [8]

6.Methodology

To carry out the research, a group of 450 people will be selected. The data of age,height,weight,heart health,amount of cholesterol or congestive heart issues as assessed by echodiagram, blood tests, stress levels and electrocardiogram of the patients will be collected.The whole-genome genotyping for all the people will be done using manual DNA extraction method. For the next step, sequencing can be performed in Illuma. Once the sequencing is done, genomic annotation will be done. In this way, the identification of functional genomic segments should be done.[5]Single marker linear regression will be performed in PLINK using the square root of warfarin dose. To improve the dose prediction *VKORC1* and *CYP2C9* will also be variables. For the implementation of RFR,a model will be installed that would assess the dosage in concentration in accordance to milligrams. The independent variable will be the *VKORC1* and *CYP2C9* genotype along with height,weight and health factors. This will provide insights into which genes are most influential in determining dosage concentration.[7] The timeline will be 3 months for assessing and repetition of the tests.

Results

Independent Variable- *VKORC1* and *CYP2C9* genotype along with height,weight and health factors.

Dependent Variable- milligrams of dosage expressed in percentage.

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